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On a Remez-type inequality for
trigonometric polynomials

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Abstract

We obtain a Remez-type inequality for a trigonometric polynomial Q_n of degree at most n with real coefficients

$$\|Q_n\|_{C((-\pi, \pi])} \leq (1/2)(2/\sin(\lambda/4))^{2n} \|Q_n\|_{C(E)}, \quad \lambda \in (0, 2\pi],$$

where $E \subseteq (-\pi, \pi]$ is a measurable set with $|E| \geq \lambda$. This estimate is asymptotically sharp as $\lambda \rightarrow 0+$, that is, for the best constant $C_{n, \mathbb{R}}(\lambda)$ in this inequality, $C_{n, \mathbb{R}}(\lambda) = (1/2)(8/\lambda)^{2n}(1 + o(1))$. We also extend this result to polynomials with complex coefficients.

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1. Introduction

Throughout the note n is a fixed natural number. Let $\mathcal{T}_{n, \mathbb{R}}$ or $\mathcal{T}_{n, \mathbb{C}}$ be the set of all trigonometric polynomials $Q_n(t) = \sum_{k=0}^n (a_k \cos kt + b_k \sin kt)$ of degree at most n with real or complex coefficients, respectively. Let us set

$$C_{n, \mathbb{R}}(\lambda) := \sup_{|E| \geq \lambda} \sup_{Q_n \in \mathcal{T}_{n, \mathbb{R}}, \|Q_n\|_{C(E)} \leq 1} \|Q\|_{C((-\pi, \pi])}, \quad \lambda \in (0, 2\pi], \quad (1.1)$$

where $\|f\|_{C(E)} := \sup_{t \in E} |f(t)|$ and the second supremum in (1.1) is taken over all measurable sets $E \subseteq (-\pi, \pi]$ of the Lebesgue measure $|E| \geq \lambda$. In other words, $C_{n, \mathbb{R}}(\lambda)$ is the best constant

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in the inequality $\|Q_n\|_{C(-\pi, \pi]} \leq C\|Q_n\|_{C(E)}$, $|E| \geq \lambda$, $Q_n \in \mathcal{T}_{n, \mathbb{R}}$. Replacing $\mathcal{T}_{n, \mathbb{R}}$ with $\mathcal{T}_{n, \mathbb{C}}$ in (1.1), we denote the corresponding constant by $C_{n, \mathbb{C}}(\lambda)$.

Constant $C_{n, \mathbb{R}}(\lambda)$ is unknown, while the corresponding constant for algebraic polynomials was found by Remez [11]. In this note we investigate the asymptotic behavior of $C_{n, \mathbb{R}}(\lambda)$ as $\lambda \rightarrow 0+$. In addition, we show that $C_{n, \mathbb{R}}(\lambda) = C_{n, \mathbb{C}}(\lambda)$.

A lower estimate

$$C_{n, \mathbb{R}}(\lambda) \geq A_n(\lambda) := (1/2)(\tan^{2n}(\lambda/8) + \cot^{2n}(\lambda/8)) \quad (1.2)$$

is well known (see [8, p. 232], [4,7,12, p. 90]). Here,

$$A_n(\lambda) = (1/2)(8/\lambda)^{2n}(1 + o(1)), \quad \lambda \rightarrow 0+, \quad (1.3)$$

and

$$A_n(\lambda) = 1 + n^2(2\pi - \lambda)^2/8 + O((2\pi - \lambda)^4) = \exp(n^2(2\pi - \lambda)^2/8)(1 + o(1)), \quad \lambda \rightarrow 2\pi - . \quad (1.4)$$

Asymptotic (1.3) is trivial and (1.4) is established in [10].

Upper estimates for $C_{n, \mathbb{R}}(\lambda)$ were obtained in several publications. In particular, Erdélyi [4] showed that

$$C_{n, \mathbb{R}}(\lambda) \leq \exp(A_n(2\pi - \lambda)), \quad \lambda \in [3\pi/2, 2\pi],$$

where A is an absolute constant. An estimate $A < 2$ was found in [7]. Recently Nursultanov and Tikhonov [10] established the inequality

$$C_{n, \mathbb{R}}(\lambda) \leq 1 + 2 \tan^2(n(2\pi - \lambda)/4), \quad \lambda \in (2\pi - \pi/n, 2\pi]. \quad (1.5)$$

It follows from (1.2), (1.4) and (1.5) that estimate (1.5) is asymptotically sharp as $\lambda \rightarrow 2\pi -$, that is,

$$C_{n, \mathbb{R}}(\lambda) = 1 + n^2(2\pi - \lambda)^2/8 + O((2\pi - \lambda)^4), \quad \lambda \rightarrow 2\pi -, \quad (1.6)$$

(see [10]). Nazarov [9] proved the inequality

$$C_{n, \mathbb{R}}(\lambda) \leq A_1(A_2/\lambda)^{2n}, \quad \lambda \in (0, 2\pi], \quad (1.7)$$

where $A_1 \leq 1$ and $A_2 \leq 87$. Estimates $A_1 \leq \sqrt{2}$ and $A_2 \leq 17$ were obtained in [7]. More results and references can be found in [5,3,7,1,6].

In this note we establish the following Nazarov-type result.

Theorem 1. (a) *The following relations hold:*

$$C_{n, \mathbb{R}}(\lambda) = C_{n, \mathbb{C}}(\lambda) \leq (1/2)(2/\sin(\lambda/4))^{2n}, \quad \lambda \in (0, 2\pi]. \quad (1.8)$$

(b) *Estimate (1.8) is asymptotically sharp as $\lambda \rightarrow 0+$, that is,*

$$C_{n, \mathbb{R}}(\lambda) = (1/2)(8/\lambda)^{2n}(1 + o(1)), \quad \lambda \rightarrow 0+. \quad (1.9)$$

Remark 2. Better estimates $A_1 \leq 1/2$ and $A_2 \leq 4\pi$ in (1.7) follow immediately from (1.8). In addition, we note that asymptotics (1.6) and (1.9) support the conjecture in [5,7] that $C_{n, \mathbb{R}}(\lambda) = A_n(\lambda)$.

2. Proof of Theorem 1

Asymptotic (1.9) follows immediately from (1.2), (1.3) and (1.8). The proof of relations (1.8) is based on two lemmas.

The first of them actually states that in the definition of $C_{n,\mathbb{C}}(\lambda)$ we can restrict ourselves to trigonometric polynomials with only real zeros. In the real case, various versions of this result were proved in [4, Proposition 1] by a variational method and in [7, Lemma 3] by an alternance method. We use a different approach to prove the following result.

Lemma 3. *Let $\mathcal{T}_{n,\mathbb{R}}^*$ be the set of all trigonometric polynomials from $\mathcal{T}_{n,\mathbb{R}}$ with only real zeros. Then*

$$C_n(\lambda, \mathbb{C}) := \sup_{|E| \geq \lambda} \sup_{Q_n \in \mathcal{T}_{n,\mathbb{R}}^*, \|Q_n\|_{C(E)} \leq 1} |Q_n(\pi)|, \quad (2.1)$$

Proof. It is easy to see that $\|Q_n\|_{C((-\pi, \pi))}$ in the definition of $C_n(\lambda, \mathbb{C})$ can be replaced with $|Q_n(\pi)|$. Let $E \subseteq (-\pi, \pi]$, $|E| \geq \lambda$, and $Q_n \in \mathcal{T}_{n,\mathbb{C}}$ satisfy the conditions $Q_n(\pi) \neq 0$ and $\|Q_n\|_{C(E)} = 1$. Then by the substitution $y = \tan(t/2)$, the polynomial Q_n can be represented as a rational function

$$Q_n(t) = R_{2n}(y) = (1 + y^2)^{-n} P_{2n}(y) = A(1 + y^2)^{-n} \prod_{k=1}^{2n} (y - a_k - ib_k), \quad (2.2)$$

where P_{2n} is a polynomial with zeros $a_k + ib_k$, $1 \leq k \leq 2n$, and $A \in \mathbb{C}$, $a_k \in \mathbb{R}$, $b_k \in \mathbb{R}$, $1 \leq k \leq 2n$. On the other hand, any rational function R_{2n} defined by (2.2) belongs to $\mathcal{T}_{n,\mathbb{C}}$ as a function of t . Next we see that $\|R_{2n}\|_{C(E_1)} = 1$, where $E_1 := \{\tan(t/2) : t \in E\}$, and, in addition, $|Q_n(\pi)| = |A|$. Let us set

$$Q_n^*(t) = R_{2n}^*(y) := |A|(1 + y^2)^{-n} \prod_{k=1}^{2n} (y - a_k).$$

Then $Q_n^* \in \mathcal{T}_{n,\mathbb{R}}^*$, $|Q_n(\pi)| = |A|$, and

$$\|Q_n^*\|_{C(E)} = \|R_{2n}^*\|_{C(E_1)} \leq \|R_{2n}\|_{C(E_1)} = 1.$$

Therefore, the trigonometric polynomial $Q_n^{**}(t) := Q_n^*(t)/\|Q_n^*\|_{C(E)}$ has only real zeros and it satisfies the properties $\|Q_n^{**}\|_{C(E)} = 1$ and $|Q_n^{**}(\pi)| \geq |Q_n(\pi)|$. Thus (2.1) follows. $\square$

We also need a recent result by Arestov and Mendelev [2] who found the constant

$$\sigma_n(y) := \inf_{Q_{n-1} \in \mathcal{T}_{n-1,\mathbb{R}}} |\{t \in (-\pi, \pi] : |y \cos nt - Q_{n-1}(t)| \geq 1\}|. \quad (2.3)$$

We rewrite (2.3) in the following form.

Lemma 4. *For $\lambda \in (0, 2\pi]$,*

$$B_n(\lambda) := \inf_{|E| \geq \lambda} \left(\inf_{\varphi_k \in \mathbb{R}, 1 \leq k \leq 2n} \sup_{t \in E} \left| \prod_{k=1}^{2n} (2 \sin((t - \varphi_k)/2)) \right| \right) = 2 \sin^2(\lambda/4). \quad (2.4)$$

Proof. Let

$$g_{2n}(t) = g_{2n}(t, \varphi_1, \dots, \varphi_{2n}) := \prod_{k=1}^{2n} (2 \sin((t - \varphi_k)/2)) \quad (2.5)$$

be a trigonometric polynomial with only real zeros. The set of all functions (2.5) is denoted by $\mathcal{G}_{2n}$. We first note that the trigonometric polynomial

$$g_{2n}^*(t) := 2 \sin^{2n}(\lambda/4) T_n \left(\frac{\cos t - \cos^2(\lambda/4)}{\sin^2(\lambda/4)} \right) = 2 \cos nt + \sum_{j=1}^n c_j \cos(n-j)t,$$

where T_n is the Chebyshev polynomial of degree n , has only real zeros (cf. [8, p. 232], [4, 7, 12, p. 90]). Hence $g_{2n}^* \in \mathcal{G}_{2n}$. In addition, $\|g_{2n}^*\|_{C(E)} = 2 \sin^{2n}(\lambda/4)$ for $E = (-\lambda/2, \lambda/2]$. Therefore,

$$B_n(\lambda) \leq 2 \sin^{2n}(\lambda/4). \quad (2.6)$$

Next, for $h \in [0, 2)$ we set

$$\delta_{2n}(h) := \inf_{g_{2n} \in \mathcal{G}_{2n}} |\{t \in (-\pi, \pi] : |g_{2n}(t)| \geq h\}|.$$

It was shown in [2, Corollary 1 and Theorem 1] that for $h \in [0, 2)$,

$$\delta_{2n}(h) = \sigma_n(2/h) = 4 \arccos((h/2)^{1/(2n)}).$$

Hence for any $g_{2n} \in \mathcal{G}_{2n}$ and $h \in [0, 2)$,

$$|\{t \in (-\pi, \pi] : |g_{2n}(t)| \leq h\}| \leq 4 \arcsin((h/2)^{1/(2n)}). \quad (2.7)$$

Further, it is known that $B_n(2\pi) = 2$ (see [2, Eq. (1.30)]), so (2.4) holds for $\lambda = 2\pi$. Let us assume that $\lambda \in (0, 2\pi)$. Then $B_n(\lambda) < 2$, by (2.6). This shows that in the definition of $B_n(\lambda)$ in (2.4) we can restrict ourselves to sets $E \subset (\pi, \pi]$, $|E| \geq \lambda$, and polynomials $g_{2n} \in \mathcal{G}_{2n}$, satisfying the condition $\|g_{2n}\|_{C(E)} < 2$. Let $E \subset (\pi, \pi]$, $|E| \geq \lambda$, and $g_{2n} \in \mathcal{G}_{2n}$ satisfy this condition. Then setting $h = \|g_{2n}\|_{C(E)}$, we obtain from (2.7)

$$\lambda \leq |E| \leq |\{t \in (\pi, \pi] : |g_{2n}(t)| \leq \|g_{2n}\|_{C(E)}\}| \leq 4 \arcsin((\|g_{2n}\|_{C(E)}/2)^{1/(2n)}).$$

Hence

$$B_n(\lambda) \geq 2 \sin^{2n}(\lambda/4). \quad (2.8)$$

Thus (2.4) follows from (2.6) and (2.8). $\square$

Proof of Theorem 1(a). We first recall that every trigonometric polynomial $Q_n(t) = a_n \cos nt + b_n \sin nt + \dots$ from $\mathcal{T}_{n, \mathbb{R}}$ with $a_n^2 + b_n^2 \neq 0$ can be represented in the form $Q_n(t) = A \prod_{k=1}^{2n} \sin((t - \varphi_k)/2)$, where $A \neq 0$ is a real constant and φ_k , $1 \leq k \leq 2n$, are zeros of Q_n . Then by Lemmas 3 and 4,

$$\begin{aligned} C_{n, \mathbb{R}}(\lambda) &= C_{n, \mathbb{C}}(\lambda) = \sup_{|E| \geq \lambda} \sup_{Q_n \in \mathcal{T}_{n, \mathbb{R}}} |Q_n(\pi)| / \|Q_n\|_{C(E)} \\ &= \sup_{|E| \geq \lambda} \left(\sup_{\varphi_k \in \mathbb{R}, 1 \leq k \leq 2n} \left| \prod_{k=1}^{2n} \cos(\varphi_k/2) \right| \middle/ \sup_{t \in E} \left| \prod_{k=1}^{2n} \sin((t - \varphi_k)/2) \right| \right) \\ &\leq 2^{2n-1} \sin^{-2n}(\lambda/4). \end{aligned}$$

Hence (1.8) follows. $\square$

Remark 5. Note that an analogue of (1.7) holds in a more general case of generalized exponential polynomials [9], while the method used in this paper to prove (1.8) does not seem to work for these polynomials.

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